



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

ASSIGNMENT No: 8

TITLE: Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.

Conclusion:

The Java program was successfully implemented to demonstrate the use of the super keyword in inheritance. The super keyword was used to access parent class variables, constructors, and methods from the child class, ensuring proper constructor chaining and code reusability. The Employee-Manager and Vehicle Insurance System examples helped in understanding how super simplifies inheritance and supports effective object-oriented programming in Java.

Exercise

1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.

Code:

```
1  class Employee {
2      String name = "Rahul";
3      int empId = 101;
4  }
5
6  class Manager extends Employee {
7
8      void display() {
9          System.out.println("Employee Name: " + super.name);
10         System.out.println("Employee ID: " + super.empId);
11     }
12
13     Run | Debug
14     public static void main(String[] args) {
15         Manager m = new Manager();
16         m.display();
17     }
18 }
```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\cl404_30\Desktop\pratice> & 'C:\Program Files\Java\jdk1.8.0_260\bin' 'Manager'
Employee Name: Rahul
Employee ID: 101
PS C:\Users\cl404_30\Desktop\pratice>
```

2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

Code:

```

1  class Vehicle {
2
3      String name = "Car";
4      String model = "Toyota";
5
6      // Inner Class
7      class VehicleDetails {
8          void display() {
9              System.out.println("Vehicle Name: " + name);
10             System.out.println("Vehicle Model: " + model);
11         }
12     }
13
14     Run | Debug
15     public static void main(String[] args) {
16
17         Vehicle v = new Vehicle();
18
19         // Inner Class Object
20         VehicleDetails vd = v.new VehicleDetails();
21         vd.display();
22
23         // Anonymous Class
24         Runnable action = new Runnable() {
25             public void run() {
26                 System.out.println(x: "Vehicle is Running...");
27             }
28         };
29
30         action.run();
31     }

```

PROBLEMS 1 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\cl404_30\Desktop\pratice> & 'C:\Program Files\Java\jdk1.8.0_25\bin\java.exe' -cp 'C:\Program Files\Java\jdk1.8.0_25\bin\pratic
Vehicle Name: Car
Vehicle Model: Toyota
Vehicle is Running...
PS C:\Users\cl404_30\Desktop\pratice>

```